**Slide 1. Purpose / pace:** Opening; about 30 seconds.

**Talking points:** The motivating question is how to estimate images, motion, and depth when the sensor reports changes rather than conventional image frames. The mathematical thread is the interaction of observation null spaces, regularization, and identifiability.

**Explanation / transition:** The talk first introduces the sensor without assuming a computer-vision background, then develops the three inverse problems. The defense date is intentionally not specified beyond the semester.




**Slide 2. Purpose / pace:** Motivation; about 60 seconds.

**Talking points:** Left is a conventional reference image of the outdoor scene. Center is a 50 ms sample of the event measurements used by the reconstruction: blue marks positive-only pixels, orange negative-only pixels, and purple pixels with both polarities. Right is our event-derived reconstruction, using a longer observation interval. The arrow represents events feeding the estimator; the reference supplies visual context. Events determine temporal changes, and the model selects a brightness baseline and how to balance observations with regularization.

**Explanation / transition:** All panels use the raw left DAVIS geometry, $346 \times 260$ pixels. The 50 ms display interval ends at reconstruction time 1506118045.9162312 s and contains 23,780 events. The nearest reference is 4.672 ms later. Reconstruction uses the full 500,000-event input spanning about 0.846 s; the center panel displays a subset of that input. Event color strength is logarithmic in count and saturates at four events per pixel. The reference uses a sequence-wide min/max mapping; reconstruction uses one affine mapping across its 20-frame sequence, then clips to $[0, 1]$. This displays a latent event-derived state rather than calibrated absolute brightness. The dissertation asks what observations determine, what penalties select, and what guarantees survive spatial or binocular coupling.




**Slide 3. Purpose / pace:** Upfront route; about 45 seconds.

**Talking points:** First introduce what the sensor reports and how we represent those observations. Spend most of the talk on reconstructing an image over time from changes. Then develop the main optical-flow theorem, give a brief stereo-depth summary, and finish with the contributions and next questions.

**Explanation / transition:** The same question links these stages: what do the observations determine, and what does the model supply? The footer tracks our position along this route. Start with the sensor itself: each pixel reports its own changes.




**Slide 4. Purpose / pace:** Positioning; about 90 seconds.

**Talking points:** For image reconstruction, Rebecq and colleagues use a recurrent neural network trained on simulated events. Zhang, Yezzi, and Gallego use optical flow to formulate a linear image inverse problem, with classical or learned image regularization. Antil and Sayre combine events with conventional frames in a bilevel problem that estimates event-threshold parameters. Our published temporal work reconstructs each pixel independently from event increments. I develop the temporal kernel representation and activity-based penalties, then analyze the fixed quadratic solve: what the observations determine, how regularization selects a baseline, and how perturbations affect the solution.

**Explanation / transition:** For flow, the starting points are Horn and Schunck's data-plus-smoothness formulation and Benosman and colleagues' event-timestamp geometry. I formulate the time-surface, voxel-projection, and structure-tensor models in a common quadratic family and identify conditions that give existence, uniqueness, and stability. Classical variational flow already has well-posedness theory, including Weickert and Schnörr's framework; this dissertation supplies the event-derived coefficient constructions and their explicit identifiability and perturbation hypotheses. Begin the models by explaining what the sensor actually reports.

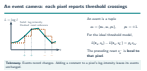



**Slide 5. Purpose / pace:** Accessibility primer; about two minutes. Assume no event-camera background.

**Talking points:** Point to the left panel: a conventional camera samples every pixel at the same frame times, whether or not anything changes, and averages over its exposure. Point to the right panel: the same moving bright bar produces a brightening event when its leading edge reaches a pixel and a darkening event when its trailing edge leaves. This is why event displays often look like moving outlines: a boundary passing across a pixel changes its brightness, while a uniform region undergoing the same translation produces little activity. The sensor responds to temporal brightness change; illumination changes can also trigger events. The survey figures quoted on the slide are the standard sensor-class numbers from Gallego et al. (TPAMI 2022), not measurements made in this dissertation.

**Explanation / transition:** Give the committee the one idea the whole talk uses: events record when the log-brightness at a pixel changed by a fixed amount, so the data determine changes and leave the starting brightness free. Then turn to the next slide for the exact threshold rule. If asked about hardware: the dissertation's MVSEC, office, and cup data come from DAVIS346 sensors, and the orbital data from the Falcon Neuro instrument on the ISS.

An event camera: each pixel reports threshold crossings




**Slide 6. Purpose / pace:** Sensor primer; 90 seconds.

**Talking points:** Follow the solid curve. A positive event is emitted after an upward change of one threshold from the most recent reset; a negative event follows a downward change. Different pixels have independent reset times, so the observation times are asynchronous. Neuromorphic here refers to the change-triggered sensing architecture; these estimators do not require a neural network.

**Explanation / transition:** The equality is an ideal noiseless threshold model. Real threshold uncertainty, missed events, refractory effects, and overshoot create discrepancies. The diagram is conceptual, with arbitrary axes and no claim about measured latency. A stationary bright pixel can produce no events: zero activity is not zero intensity.

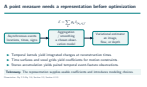



**Slide 7. Purpose / pace:** Accessibility primer; 90 seconds. Introduce the three estimated outputs.

**Talking points:** Left: temporal image reconstruction. Middle: optical-flow estimate, with color encoding direction and displacement magnitude at pixels that recorded events. Right: stereo depth. All three show the same indoor flying 2 scene. Present the panels as the three estimation targets. Explain the brightness baseline, motion along an edge, and stereo correspondence.

**Explanation / transition:** If asked which methods: B-spline kernel with sigmoid weights; contrast maximization, whose nonconvex objective lies outside the quadratic family analyzed later (existence result in backup); weighted least squares. Reconstruction: 47.690 s. Flow: 47.592–47.719 s; color magnitude represents displacement over this window. White flow pixels have zero events: the mask shows event support, not confidence. WLS: center 47.598 s, 250 ms window. Different windows and stereo rectification permit scene-level comparison, with separate pixel geometries. Depth colors clip to 0.5–8 m; white masks nonfinite values and the maximum finite depth. Reconstruction grayscale uses a histogram/percentile scale. Exact timestamps, normalization, and provenance are in the adjacent metadata file.

A point measure needs a representation before optimization




**Slide 8. Purpose / pace:** Representations; 60 seconds.

**Talking points:** A finite signed atomic measure has no ordinary density that can simply be differentiated and squared in an integral energy. Convolution, interpolation, and binning produce the quantities the models use. Raw events do have distributional derivatives; the issue is that the proposed squared pointwise residuals cannot be formed directly from them.

**Explanation / transition:** Distinguish physical intensity I, log-intensity L, event activity, and latent state f. In Chapter 2, f approximates threshold-scaled log-intensity changes; in Chapter 4, preprocessing makes f an uncalibrated event-feature profile. This distinction prevents a physical interpretation from being silently transferred between chapters.



**Slide 9. Purpose / pace:** Temporal observation model; 75 seconds.

**Talking points:** Suppress the pixel index to keep the notation readable. Integrating a unit-mass kernel across a bin assigns all or part of each event to that bin. Delta binning has no spread; Gaussian, rectangular, and spline kernels distribute influence over time. The entries of A are unscaled increments, not derivatives divided by a time step.

**Explanation / transition:** If Af=0, every consecutive pair of components is equal, hence f is constant. Conversely A1=0. Under ideal threshold calibration the observed increments approximate changes in L/c_th, with kernel-boundary, overshoot, and between-event endpoint errors. No absolute frame is used to set the baseline. Scene terms: each pixel's brightness history is rebuilt on its own; a pixel without events stays at the reference level.




**Slide 10. Purpose / pace:** Kernel choice; 75 seconds.

**Talking points:** The four kernels are the ones evaluated in Chapter 2: delta (binning), truncated and renormalized Gaussian, B-spline, and rectangular. Unit mass means each event contributes exactly its sign in total; translation invariance means the bin integrals are a convolution; compact support means each event touches finitely many bins. The sum identity is exact: summing the bin integrals telescopes to the integral over the whole window, which is one when the event's full support lies in the window.

**Explanation / transition:** The plotted widths are matched for display only; the evaluated parameters differ by kernel (Ch. 2 bump sweep). If asked which kernel is best: the indoor and outdoor rankings differ (discussion page on kernel performance), and the uniqueness theorem holds for every kernel because it depends only on **A** and positive weights. Transition: the kernels also give a continuous-time interpretation of the model.



**Slide 11. Purpose / pace:** Continuous-time connection; 90 seconds.

**Talking points:** Mollify each event at scale h; the integrated signal is a smooth staircase that converges in L1 to a pure jump function as h goes to zero. Its distributional derivative is the signed event measure, so the cumulative signal solves a measure-valued initial-value problem. Integrating over a reconstruction bin recovers exactly the observed net polarity, which is what the rows of A f approximately match.

**Explanation / transition:** The cumulative signal g is the unregularized event integral: it is neither the physical log-intensity, which needs an absolute reference, nor the Tikhonov minimizer. A jump function with a nonzero jump is in BV but not in H1, which is why the method works with integrated bins rather than pointwise derivatives of raw events. The measure-valued limit is also on the discussion page on the jump primitive. Transition: since the data fix only increments, a penalty must choose the level, which is the next theorem.

A regularized estimate is uniquely selected



**Slide 12. Purpose / pace:** Model, computation, and theorem; about 90 seconds.

**Talking points:** The objective balances fidelity to observed changes against weighted magnitude. For one pixel, differentiate the quadratic to obtain $\mathbf{K}_{\mathrm{time}}\mathbf{f} = \mathbf{A}^{\top}\mathbf{y}$. Form that pixel's matrix and right-hand side, solve the linear system numerically, and retain the resulting temporal profile. The implementation parallelizes these independent pixel solves and assembles the profiles into reconstructed frames. Its MATLAB routines use matrix left division inside a parallel pixel loop. Strictly positive fixed weights make $\mathbf{K}_{\mathrm{time}}$ positive definite and select one global minimizer.

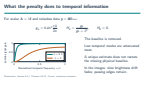
**Explanation / transition:** Premultiply the normal equations by the constant vector and use A times one equals zero to obtain the weighted zero-mean identity. The magnitude penalty selects the baseline while also changing fitted increments. Under a fixed Gaussian observation model with unit covariance and latent prior precision Lambda transpose Lambda, the same minimizer is the MAP estimate. That interpretation uses fixed positive weights; data-derived weights give a plug-in fixed-weight model. With scalar regularization tending to zero, the solution tends to the minimum-norm exact fit, which sets the constant null mode to zero. Scene terms: intervals with larger weights set the reference level.



**Slide 13. Purpose / pace:** Proof; 90 seconds.

**Talking points:** Walk through the quadratic form. Both terms are nonnegative, and the strictly positive diagonal penalty supplies a lower bound in every direction. Finite-dimensional coercivity gives existence; the positive-definite Hessian gives uniqueness. Subtracting the two normal equations gives the exact perturbation identity.

**Explanation / transition:** The displayed full-space bound uses $\left\|\mathbf{K}^{-1}\right\|_2 \leq \lambda_{\min}^{-2}$. A sharper operator bound is $\left\|\mathbf{K}^{-1}\mathbf{A}^\top\right\|_2 \leq 1/(2\lambda_{\min})$: from $\mathbf{A}^\top\mathbf{A} \preceq \mathbf{K} - \lambda_{\min}^2 \boldsymbol{I}$, the squared singular-value bound is at most $\max_{t \geq \lambda_{\min}^2}(1/t - \lambda_{\min}^2/t^2) = 1/(4\lambda_{\min}^2)$. The displayed estimate is a worst-case bound; the reported condition numbers describe the evaluated configurations. Scene terms: one spurious or missed event gives $\|\delta\mathbf{y}\| \leq 1$ (nonnegative unit-mass kernels), so the bound limits how far one noise event moves a history.



**Slide 14. Purpose / pace:** Spectral interpretation; 75 seconds.

**Talking points:** The first-difference Gramian is the Neumann-chain Laplacian. Its constant eigenvector has eigenvalue zero. Expanding a noiseless latent signal in the eigenbasis turns reconstruction into multiplication by g/(g+lambda^2). These are latent-signal gains; the gain from noise in the difference observations is a different expression in the backup.

**Explanation / transition:** The spectral decomposition is proved in Chapter 4 and reused here for stereo. The plotted smooth curves interpolate the finite-chain modal gains. Only omega_k=pi k/n are exact finite-grid modes. This magnitude penalty attenuates low temporal frequencies most, whereas the temporal kernels may average rapid changes. Scene terms: slow brightness drift fades and passing edges remain, hence the edge-enhanced look.

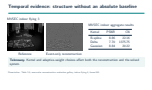
How are the temporal weights chosen?




**Slide 15. Purpose / pace:** Adaptive regularization choice; 90 seconds.

**Talking points:** The event activity here is the absolute value of the signed accumulation in one temporal bin. Larger activity produces a smaller magnitude penalty. Sigmoid was selected as an empirical fidelity/conditioning compromise, not as an optimized or certified best lambda. Its reported CN diagnostic is lower than the alternatives outdoors, while inverse proportional has a lower reported CN indoors.

**Explanation / transition:** The other two rules and their exact formulas are in the discussion page "Adaptive temporal weights are fixed in the theorem". The reported outdoor CN diagnostics are 29.60 for sigmoid, 81.50 for inverse proportional, and 100.74 for exponential decay. For any fixed positive weights, the temporal quadratic has a unique minimizer and the stability bound applies at those weights. The exact comparison identity $K_1(f_1 - f_2) = A^T(y_1 - y_2) - (\mathbf{\Lambda}_1^2 - \mathbf{\Lambda}_2^2)f_2$ shows the extra weight-change term when data-dependent weights are recomputed; this direct identity does not establish a full adaptive stability theorem. Scene terms: quiet intervals get the largest weights and set the reference level; busy intervals get the smallest and set the worst-case noise bound.



**Slide 16. Purpose / pace:** Temporal results; 60 seconds.

**Talking points:** The pictures show a representative indoor flying 4 reference and event-only reconstruction with sigmoid weights. The table aggregates the indoor kernel comparison. PSNR measures reference-image similarity; a higher value means closer agreement. CN measures conditioning; a lower value indicates better conditioning. B-spline and Gaussian give similar PSNR and much lower CN than Delta in this indoor comparison.

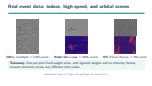
**Explanation / transition:** The reconstruction recovers scene structure, while the magnitude penalty selects the absolute baseline. The outdoor kernel comparison has a different ordering and appears in the backup. These results establish the observed tradeoff for the tested data and parameter choices. Display normalization is a visualization choice; the theorem's uniqueness and stability guarantees apply with the positive weights fixed.




**Slide 17. Purpose / pace:** Relationship and comparison; 90 seconds.

**Talking points:** In Anti–Sayre, the outer variable is event threshold parameter $z$, not a learned regularization weight. Their problems are $u(z) = \arg\min_u \frac{1}{2}\|Au - b(z)\|^2 + \frac{\lambda_2}{2}\|u\|^2$ and $\min_z \frac{1}{2}\|u(z) + g(z) - d\|^2 + \frac{\lambda_1}{2}\|z\|^2$, with $d$ log frames, $g$ an event-dependent exposure term, and $b = A[d - g(z)]$. Our Chapter 2 rule computes diagonal magnitude weights from event activity and then solves a fixed convex quadratic.

**Explanation / transition:** The table reproduces the dissertation's indoor and outdoor averages: PSNR/CN are 15.37/39755 versus 10.85/25.51 indoors, and 8.41/39755 versus 8.71/27.10 outdoors, for bilevel versus event-only. The methods receive different inputs, including absolute intensity frames for the comparator, so this is descriptive rather than controlled evidence of superiority. The paper analyzes existence and conditional local uniqueness for its bilevel problem; our theorem establishes strict convexity and uniqueness for each fixed positive $\mathbf{\Lambda}$. These guarantees and the lambda coefficients have different roles.




**Slide 18. Purpose / pace:** Qualitative real-data results; 90 seconds.

**Talking points:** Office: a handheld DAVIS346 in an office; the desk objects and the person are recognizable. Cup: a bullet strikes a ceramic cup from the left; the event stream resolves the fracture and spray, which a standard-rate frame camera would blur. ISS: the Falcon Neuro event cameras on the International Space Station viewing the eastern coastline of Honduras; land–water boundaries and clouds are visible although the scene changes slowly and the events are sparse. Bottom row of the second and third columns is a false-color display of the same reconstruction.

**Explanation / transition:** These are qualitative demonstrations; the quantitative MVSEC comparison is on the preceding pages. Grayscale and false color are display choices, so do not describe them as calibrated brightness. If asked why the results look like edge-enhanced images: low temporal modes are attenuated most (spectral slide), and the baseline is selected by the penalty, so the reconstruction emphasizes changes. Transition: the next slide lifts these per-pixel constraints to a scene that moves relative to the camera.

Shared motion couples temporal profiles to one scene



**Slide 19. Purpose / pace:** Shared-scene temporal model; 75 seconds.

**Talking points:** Here $\theta$ contains static scene values only. The projections $P_k$ encode fixed, known motion and map that scene to each frame; $A$ differences the profiles. The nullspace contains parameters whose projected profiles agree across frames. Constants are included because each projection is row-normalized; other invariant scene modes may also occur. This resembles flow's aperture ambiguity in that observations leave directions uncontrolled, but proves no motion ambiguity for SAR.

**Explanation / transition:** Here lambda is the unsquared ridge coefficient in the normal equations. For this fixed-motion linear model, the stability estimate is $\|\delta\theta\| \le \|B_{\mathrm{SAR}}\| \, \|\delta b\| \, / \lambda$. The theorem does not say every SAR nullspace is strictly larger than constants; that depends on P.



**Slide 20. Purpose / pace:** Define the aperture problem visually; about 90 seconds.

**Talking points:** Optical flow is apparent velocity in the image plane. An aperture simply means a restricted observation window. First look at the circular windows: the moving stripes look alike. Revealing the full patterns shows three different translation directions. A local straight edge tells us its speed perpendicular to the edge; motion along its tangent stays free. These publisher animations illustrate motion perception. They are conceptual demonstrations, rather than measured event-camera data.

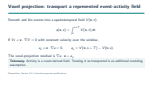
**Explanation / transition:** The two animations are embedded on this slide. Play the masked panel first using its forward-play control, then play the revealed panel; each loops at the original rate. First-frame posters preserve the visual comparison in static viewers and print. Use a PDF viewer with animation support and JavaScript enabled, as documented with the assets. Under steady illumination, different velocities with the same normal component produce the same local brightness changes and ideal events. Smoothness couples neighboring velocity estimates, while additional observed directions constrain the component left free by a single edge. Next derive the crossing-time constraint that expresses this ambiguity mathematically. Scene-terms summary (spoken): velocity at every pixel; events give speed perpendicular to each edge; neighbors move alike; requirement is edges in two orientations somewhere.




**Slide 21. Purpose / pace:** Geometric derivation; 60 seconds.

**Talking points:** This chain-rule argument is accessible without any sensor-specific machinery. Follow a material point on the translating edge and differentiate the identity saying it stays on the zero level set.

**Explanation / transition:** The coefficient bounds used later apply to fitted or mollified maps. The timestamp-regression result assumes a differentiable crossing-time map with Lipschitz Hessian, but raw last-timestamp maps do not automatically satisfy those assumptions. A timestamp may belong to an old edge or multiple crossings; the variational theorem bounds usable coefficient perturbations under uniform identifiability. Scene terms: $|\nabla \mathcal{T}|$ is slowness; an edge taking 10 ms per pixel moves at 100 px/s normal to itself.




**Slide 22. Purpose / pace:** Voxel model; 75 seconds.

**Talking points:** A voxel grid adds a time axis to the image plane and stores represented event contributions in time bins. The activity field is not an intensity image: a bright stationary scene can have zero activity. For an exactly transported density with constant velocity, integrate the PDE over the sliding window; this gives the displayed transport identity exactly, even for a large displacement.

**Explanation / transition:** Finite temporal projection changes the profile by averaging its translates. The evaluated spatial bin mean and temporal OLS slope need not be derivative-consistent partners, and asymmetric smoothing can add mismatch. The well-posedness theorem only needs fixed bounded coefficient fields; an accurate physical-velocity interpretation also needs the transport approximation.




**Slide 23. Purpose / pace:** Structure tensor; 75 seconds.

**Talking points:** Expand the quadratic: v^T Jtilde v + 2 j_t^T v + J_tt. The last term is independent of velocity, so the spatial block is the data curvature and the temporal cross terms give the forcing. The temporal component of v_3 is fixed to one; this is an ordinary least-squares transport model, not a total-least-squares eigenvector model.

**Explanation / transition:** Nonnegative pooling preserves positive semidefiniteness, since z^T J z is an integral of nonnegative squared directional derivatives. A locally planar event surface is a model assumption, and smoothing scale limits temporal detail. The theorem will not require every pixel's spatial block to be positive definite.

Three representations, one quadratic variational problem




**Slide 24. Purpose / pace:** Unification; 60 seconds.

**Talking points:** The energies agree with the representation-specific residual energies up to an overall positive factor and a constant independent of flow. The common space is H^1(Omega; R^2), and the Dirichlet term couples neighboring velocities.

**Explanation / transition:** The notation B here is A_data in the dissertation, and alpha is alpha_smooth. The change avoids using A for two different operators on consecutive chapters. The common analysis is about fixed coefficients. A good mathematical estimator can still encode a poor representation of physical motion. Scene terms: **B** records which motion directions the edges measure and how strongly; $b$ the measured values.




**Slide 25. Purpose / pace:** Central theorem; 90 seconds.

**Talking points:** Classical quadratic optical flow already has well-posedness theory, including Weickert and Schnörr. The contribution here is one analysis for three event-derived coefficient constructions, with explicit identifiability and perturbation hypotheses. Read the hypotheses and interpret each. Connectedness makes the kernel of the gradient consist of global constants. The positive regularization coefficient controls spatial variation. Bounded data curvature and L2 forcing make the weak forms bounded. Positive definiteness of the integrated B controls constant fields that smoothness cannot see.

**Explanation / transition:** B may vanish on open subsets while its integral remains positive definite. In these observation models the residual forcing b also vanishes there, and smoothness propagates information from observed regions. For time surfaces, no nonzero vector can be perpendicular to all measured gradients almost everywhere. The condition need not hold pointwise. Scene terms: edges in at least two orientations somewhere in the image.



**Slide 26. Purpose / pace:** Main proof; 2 minutes.

**Talking points:** The gradient seminorm misses exactly the constant vector fields. Write the field as its mean plus a zero-mean part. Poincaré controls the latter. Use $\left\|\mathbf{B}^{1/2}\bar{\mathbf{v}}\right\|_{L^2} \le \left\|\mathbf{B}^{1/2}\mathbf{v}\right\|_{L^2} + \left\|\mathbf{B}^{1/2}\mathbf{w}\right\|_{L^2}$, square with the factor-two inequality, and bound the fluctuation term by $\beta_B C_P^2 \left\|\nabla\mathbf{v}\right\|_{L^2}^2$. Positive definiteness of the integrated matrix controls the mean.

**Explanation / transition:** An admissible lower bound, not an optimal constant, is $c_0 = \min\{m/(2|\Omega|), \alpha^2/(1 + C_P^2 + 2|\Omega|\beta_B C_P^2/m)\}$. Insert it into $\|\mathbf{v}\|_{H^1}^2 = |\Omega||\bar{\mathbf{v}}|^2 + \|\mathbf{w}\|_{L^2}^2 + \|\nabla\mathbf{v}\|_{L^2}^2$. Lax–Milgram gives the unique minimizer. For fixed positive $\beta_B, \alpha$, this lower bound deteriorates linearly as $m$ decreases. Scene terms: mean = common drift (camera pan), fluctuation = variation across the image; smoothness charges nothing for a pan, so the edges must fix it.



**Slide 27. Purpose / pace:** Stability proof; 75 seconds.

**Talking points:** The same coercivity constant works on both coefficient pairs because the quantitative bounds are shared. Subtract their weak equations and test with the solution difference. Cauchy–Schwarz controls the right side, and $\|\mathbf{v}_2^*\|_{H^1} \le R_b/c_0$ follows from its own weak equation. Divide by $\|\mathbf{z}\|_{H^1}$ if it is nonzero.

**Explanation / transition:** The conclusion is stability of the chosen estimator, not an error bound to ground-truth velocity. The estimate may become poor as $m \to 0$. If a density-dependent diffusion tensor is recomputed, the regularizer also changes; its $L^\infty$ perturbation must enter the bound. Scene terms: coefficient changes come from timestamp jitter, binning, and noise events; small $m$ is the stripe case on slide 29.

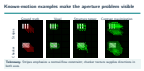


**Slide 28. Purpose / pace:** Numerical construction and theorem; about 90 seconds.

**Talking points:** Each node stores a two-component velocity. With coefficients fixed, the normal equation at node i has the local data matrix $q_i \mathbf{B}_i$ plus alpha squared times the weighted degree. Move the neighboring velocities to the right, and use their values from the previous sweep. This gives the displayed two-by-two solve. The $q_i b_i$ term carries the local observations, and the weighted neighbor sum carries the previous surrounding field. Compute every new velocity from the same old field, then replace the whole field and repeat.

**Explanation / transition:** The theorem assumes a connected graph, symmetric nonnegative edge weights, positive degrees and node factors, PSD data blocks, and a positive-definite sampled Gram matrix. For this graph operator, zero graph energy forces a constant field, and the sampled data remove that constant null direction. Write $H = D - R$, where $D$ contains the local two-by-two blocks and $R$ contains the neighbor couplings. Both $D - R$ and $D + R$ are positive definite: possible alternating constants of the signless graph form are removed by the same sampled data condition. Hence the spectrum of the similar symmetric Jacobi matrix lies in $(-1, 1)$. This proves convergence of the simultaneous schedule from every initial field. Match a representation's stencil and update schedule to these hypotheses before applying the result. Scene terms: information spreads one neighbor per sweep.

## Known-motion examples make the aperture problem visible

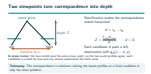



**Slide 29. Purpose / pace:** Visual evidence; 60 seconds.

**Talking points:** The object translates diagonally. Red arrows show the ground truth and green arrows the estimates, at the common scale retained from the dissertation's panels. Vertical stripe interiors provide mainly horizontal gradients. The checkerboard provides gradient information in both spatial directions. Object boundaries can supply additional information even in the stripe scene.

**Explanation / transition:** These are selected illustrative windows, not a cross-dataset method ranking. The full figure also includes time surface and other variants; its reported TS AEE equals the zero-field baseline at the reported precision. The output does not diagnose which hypothesis or extraction step fails, so do not assign a cause.

Two viewpoints turn correspondence into depth



**Slide 30. Purpose / pace:** Stereo introduction; 75 seconds.

**Talking points:** Explain rectification as a calibrated coordinate change that places corresponding rays on the same image row. Disparity is the horizontal pixel shift; similar triangles give depth as focal length times baseline divided by disparity. Farther objects have smaller disparity, so small disparity errors can create large depth errors.

**Explanation / transition:** The event-feature observations are accumulated and normalized before matching, rather than synchronous brightness images. We assume a candidate disparity is approximately valid over the observation window and both views remain visible. Occlusion, parallax, changing depth, and differing sensor responses can break profile equality. Scene-terms summary (spoken): true match sees the same changes; soft coupling; weights trust quiet intervals; best-scoring candidate gives depth.

Stereo depth in one slide: what Chapter 4 adds



**Slide 31. Purpose / pace:** Chapter 4 summary; 2 minutes. Keep to the four results and one benchmark sentence.

**Talking points:** This chapter compares candidate correspondences by reconstructing and softly coupling a temporal event-feature profile from each view. The analysis separates the unique inner reconstruction from the outer correspondence decision, then gives conditions for selection and stability under a positive winning margin. Item 1 is the Chapter 2 coercivity argument with one more nonnegative term. Item 2 is the conceptual point: after profiling out the latent profiles, the candidate cost contains a right-view monocular term that changes with the candidate, so in the two-sample example an attenuated wrong match beats the exact match. Item 3 gives a sufficient condition under which the correct candidate wins, and its stability under perturbations smaller than half the margin. Item 4 is the reweighted iteration.

**Explanation / transition:** Benchmark: each method is scored on its own valid pixels; the Piatkowska adaptation has the lowest error on Flying 1, 3, and 4. Interpret accuracy alongside scored coverage. The separation results concern unweighted candidate-local costs before aggregation; the contraction theorem applies when its displayed sufficient condition holds. Scene terms: a weaker right-view copy can outscore the true match; margin errors include noise events and sensor threshold mismatch.




**Slide 32. Purpose / pace:** Contributions; 75 seconds.

**Talking points:** Recap the mathematical content rather than the implementation chronology. The temporal null mode is selected by a penalty. Spatial smoothness and integrated directional data make the flow energy coercive. Stereo requires both a well-posed latent solve and a sufficiently separated candidate comparison. Contrast maximization uses finite-dimensional compactness because it falls outside the quadratic theory.

**Explanation / transition:** Position the originality accurately: the functional-analytic tools are classical; the dissertation organizes event-derived models under explicit hypotheses and derives the coupling, conditioning, bias, and selection consequences. The table is a useful reference when the committee asks exactly which theorem applies to the full procedure.

Next questions follow directly from the proved bounds




**Slide 33. Purpose / pace:** Research directions; 60 seconds.

**Talking points:** Tie each direction to an existing mathematical quantity: coefficient perturbation norms for flow, Lipschitz constants for weights, a winning margin for correspondence, and identifiable experimental controls for model comparison. Characterize residuals on event-active pixels before choosing a heavy-tail penalty; pooled zeros alone can inflate kurtosis.

**Explanation / transition:** A promising extension is a fixed uniformly elliptic diffusion tensor selected by a higher-level procedure, retaining a common ellipticity floor during each inner flow solve. Optimal transport between event measures is a broader option that would require explicit polarity handling, a ground cost, and a mass convention. Keep those larger changes for discussion if time is short.



**Slide 34. Purpose / pace:** Future research; 2 minutes. End the talk here before the closing slide.

**Talking points:** Learned regularization: the fixed-weight theorem gives a unique minimizer $\mathbf{f}^*(\mathbf{\Lambda}) = (\mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^2)^{-1}\mathbf{A}^\top\mathbf{y}$, a rational and hence smooth function of $\mathbf{\Lambda}$ on the positive orthant, which is what a bilevel analysis needs from its lower level. For scalar weights, a nonlocal penalty $\frac{\mu}{2}\,\mathbf{f}^\top(\mathbf{A}^\top\mathbf{A})^s\mathbf{f}$ shares the cosine eigenbasis of $\mathbf{A}^\top\mathbf{A}$, so the modal gains become $g_k/(g_k + \mu g_k^s + \lambda^2)$ and the spectral analysis carries over; the magnitude term is still needed for the constant mode. Fluids: event cameras are already used for particle-based velocimetry; there the motion field is controlled, which would test the representation-accuracy question that the flow theorem leaves to the model.

**Explanation / transition:** Shared scene: the ESAR null space contains constants and any other motion-invariant scene modes; computing it for given trajectories is a concrete linear-algebra question. Stereo: the open step is a margin result for the weighted and aggregated costs. Mention that the new GMU Neuromorphic Imaging and Digital Twins Lab (AFOSR DURIP) makes controlled acquisitions possible for the first two directions. Then go to the closing slide and stop for questions.




**Slide 35. Purpose / pace:** Close; 30 seconds.

**Talking points:** Return to the opening question. The dissertation gives a route from sparse asynchronous changes to deterministic estimators, and makes explicit the conditions under which each mathematical guarantee holds.

**Explanation / transition:** Stop here for committee questions. The appendix is organized by theorem mechanism and anticipated question; it is not part of the timed core talk.


Discussion guide

Further detail: this section includes short and context considerations, for full views see §3 (Figure 4).

Mechanism and comparison

- Alternative compression conditions — gauge
- Measure-valued nondimensional time limit
- Figure gain versus mass gain
- Relation to entropic vs. radial, and surface tension

Boundary and scaling

- Weak Allen equation and boundary condition
- Small blocks sample-spectral alignment
- Homology, cell adjacency
- Regularity design and projection mechanism
- Ferrogel energy spectrum model

- Measures conditioning
- Level-dispersal shell-dimensional perturbations
- Mesh, scattering plane
- Mesh-dependent and mesh-dependent interactions
- Environmental sensing, and aggregation

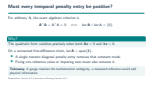
Evidence and interpretation

- Algebra procedure, coverage, and selection
- Mesh-dependent and mesh-dependent time?
- Load-stress surface and source map

**Slide 36. Purpose / pace:** Use as an appendix index.

**Talking points:** The following slides support questions rather than extend the planned talk. PDF section bookmarks and the source-map file make it possible to find their manuscript origins.

**Explanation / transition:** Each discussion slide has notes that give the proof mechanism or a suggested concise answer.




**Slide 37. Purpose / pace:** Question: why use a full positive diagonal?

**Talking points:** The main theorem's positive diagonal is a strong sufficient hypothesis with an immediate quantitative lower bound. It is not necessary for definiteness. The kernel-intersection argument shows the minimal condition. A one-point anchoring penalty gives definiteness, but its eigenvalue bound is not the same as lambda_min squared when other diagonal entries are zero.

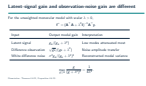
**Explanation / transition:** An explicit zero-mean parameterization can avoid penalizing the unobserved offset. It changes the estimator: the original magnitude penalty also attenuates nonconstant modes and controls noise. Choosing a gauge alone never recovers an unmeasured absolute brightness baseline.




**Slide 38. Purpose / pace:** Question: what is the continuous-time problem?

**Talking points:** A finite atomic measure has a bounded-variation cumulative primitive. For a nonzero interior atom its derivative is singular, so it cannot have an L2 weak derivative and is not H1. With finite standard bounded kernels of positive width, the smoothed driver is L2 and its primitive is H1. Arbitrary L1 alone would only guarantee W1,1.

**Explanation / transition:** The bin convention (a,b] here matches the right-continuous process exactly. Alternative half-open bins are valid with a compatible representative or an explicit no-boundary-events assumption. The difference quotient is the integrated event increment divided by the time step; the discrete A in the presentation computes increments. The limit is not a convergence theorem for the regularized minimizer as the penalty and grid vary.




**Slide 39. Purpose / pace:** Question: what kind of filter is this?

**Talking points:** Take the singular-value decomposition A=U Sigma V^T. The reconstruction operator maps left singular modes of y to right singular modes with amplitude sqrt(g)/(g+lambda squared). Multiplying by A for noiseless y=A f_true yields the latent-signal gain g/(g+lambda squared). For white independent observation noise the output covariance diagonalizes in the same right modes and its modal variance is sigma squared times the squared observation gain.

**Explanation / transition:** Differentiate g/(g+a)^2 with a=lambda squared: the derivative has sign a-g, so the continuous maximum is 1/(4a) at g=a. The discrete finite-chain maximum may be smaller because its available eigenvalues need not contain a. These formulas do not carry over unchanged to arbitrary diagonal penalties or unequal adaptive stereo weights.

Kernel performance depends on the scene and adaptive weights


Kernel performance depends on the scene and adaptive weights.

| | Kernel | Cond. | Kernel | Cond. |
|---|---|---|---|---|
| | Kernel | PSNR | Cond. | PSNR | Cond. |
| Outdoor | Bump | 9.95 | 10.15 k | 9.72 | 10.34 k |
| | Gaussian | 7.14 | 4.81 k | 6.97 | 5.14 k |
| | Box | 7.10 | 3.61 k | 6.97 | 3.86 k |
| Indoor | Bump | 4.69 | 10.15 k | 7.49 | 10.34 k |
| | Gaussian | 4.69 | 4.81 k | 7.49 | 5.14 k |
| | Box | 4.69 | 3.61 k | 7.49 | 3.86 k |

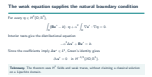
- The compared operator stays the same; represented data and adaptive penalties change.
- Same conditioning is not the same as visible reconstruction loss.
- Reference image serves (to non-identity), as absolute, linearity baseline from exams.



**Slide 40. Purpose / pace:** Question: why does a smoothing kernel change conditioning?

**Talking points:** The matrix K=A^T A+Lambda^T Lambda does not directly contain the bump kernel. However Lambda is computed adaptively from represented observations. Changing aggregation can change those weights, hence the numerical matrix. The observed condition-number difference cannot be attributed to smoothing of a fixed operator alone.

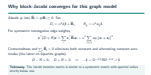
**Explanation / transition:** The table reports the dissertation's condition-number diagnostics. The outdoor ordering shows that kernel performance depends on the scene and adaptive weights. The positivity theorem is independent of the empirical ranking and gives only a sufficient worst-case bound, not a causal explanation of every numerical comparison.



**Slide 41. Purpose / pace:** Question: which boundary condition is imposed?

**Talking points:** The energy is minimized over the unrestricted H1 space, not H1_0, so natural Neumann conditions emerge from the first variation. Compactly supported smooth tests first give the interior equation. B v and b are in L2, hence the distributional Laplacian is L2, so the gradient has a weak normal trace.

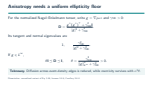
**Explanation / transition:** Returning to arbitrary H1 tests, generalized Green's identity leaves a boundary pairing with their traces. Surjectivity of the trace map makes the normal trace zero. The anisotropic form replaces grad by D grad and yields a conormal condition n^T D grad v=0. No differentiability of D is needed for its weak formulation.



**Slide 42. Purpose / pace:** Question: why does SPD suffice here for Jacobi?

**Talking points:** It does not suffice in general. The special graph form proves more. For D-R, a zero edge sum forces equality of neighbors, so all node vectors equal a constant. For D+R, it forces opposite vectors on every edge; on a bipartite graph they are alternating copies of one constant vector, while an odd cycle forces zero. The sampled PSD data blocks eliminate any nonzero constant or alternating constant.

**Explanation / transition:** D is SPD because alpha and each degree are positive. Conjugate D plus or minus R by D^-1/2; the two positivity statements bound the symmetric spectrum between minus and plus one. D^-1R is similar to it, hence its spectral radius is below one and repeated simultaneous sweeps converge. An event-triggered partial-update schedule is a separate question.



**Slide 43. Purpose / pace:** Question: can an adaptive diffusion tensor retain well-posedness?

**Talking points:** For a fixed, bounded, symmetric tensor with a common strictly positive lower eigenvalue bound, the Poincare/coercivity argument survives. Merely saying positive definite at each point is insufficient if eigenvalues approach zero. The normalized Nagel–Enkelmann form provides an explicit uniform floor when the density gradient is bounded.

**Explanation / transition:** The tensor reduces penalties on normal-direction flow variations and preserves the tangent-direction penalty. If it is recomputed after data perturbation, stability includes ||D_1-D_2||_infty. Event-density edges need not coincide with motion boundaries. The CM tensor alpha_ani squared J plus epsilon I has the opposite directional effect: it increases the normal-direction penalty.




**Slide 44. Purpose / pace:** Question: can you just replace the Laplacian by a tensor-weighted Hessian?

**Talking points:** Only for a constant tensor or when its coefficient-derivative contributions vanish. With a variable tensor the product rule supplies first-order terms. The affine test field exposes their importance particularly clearly, because its Hessian vanishes identically while its divergence-form diffusion does not.

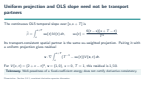
**Explanation / transition:** For merely L-infinity D, those classical coefficient derivatives may not exist. Keep the weak form integral grad eta transpose D grad u; it requires no derivative of D. A flux-consistent discretization should approximate that divergence form rather than drop terms without an error analysis.




**Slide 45. Purpose / pace:** Question: is your radius choice optimal?

**Talking points:** Differentiate the positive leading bias-squared plus variance expression, which scales as A r^4 + B r^-4. Its stationary radius obeys r^8=B/A. The resulting scaling is a guide to how timestamp noise, lattice spacing, and curvature interact.

**Explanation / transition:** The exact finite-design bound uses lattice sums. The asymptotic error has no stated sign, so the leading surrogate is not automatically an upper bound. The formula also requires positive curvature Lipschitz bound and noise, a large radius in lattice units, and inclusion in the smoothness neighborhood. Boundary truncation or sparse event support can destroy central symmetry and alter the bias.




**Slide 46. Purpose / pace:** Question: why distinguish the implementation from exact transport?

**Talking points:** Integration by parts in the OLS slope formula yields the parabolic weight w_s on the temporal derivative. Integrating the transport equation with that weight gives a correspondingly weighted spatial gradient. Using a uniform mean on one side and an OLS slope on the other generally gives a residual mismatch.

**Explanation / transition:** Uniform and OLS weights share zeroth and first temporal moments, so transported polynomials through degree two have zero mismatch. A difference between their averages for a quadratic profile does not prove a nonzero transport residual. The cubic example supplies an actual derivative mismatch. This is a representation-coefficient error that the abstract stability theorem can address if its norm and identifiability hypotheses are controlled.



**Slide 47. Purpose / pace:** Question: is small-lambda ill-conditioning unavoidable?

**Talking points:** For this scalar unweighted system the right-hand side has zero mean in each view since A1=0, and the matrix preserves that subspace. For lambda>0 the solution lies there. Removing the constant DCT mode makes the smallest eigenvalue g_1+lambda squared instead of lambda squared, and that is positive even at lambda=0.

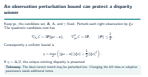
**Explanation / transition:** At fixed temporal dimension and coupling this gives a finite bound uniform over lambda>=0. However g_1 is asymptotic to pi squared/n squared as the unscaled temporal chain is refined, so the uniform bound grows quadratically. At a fixed positive lambda the actual restricted condition number remains finite under this unscaled refinement. The full system is singular at lambda=0 and unstructured rounding errors need not respect the mean-zero space. For positive fixed diagonal weights bounded by $W_{\max}$, $\kappa_2(\mathbf{M}_W) \le (4W_{\max} + \mu_{\max} + 2\gamma)/\mu_{\min}$, where $\mu_{\min}, \mu_{\max}$ are the extreme eigenvalues of $\mathbf{\Lambda}^2$.




**Slide 48. Purpose / pace:** Question: can the monocular bound be weakened?

**Talking points:** Complete the square in e using e_0=P^-1 S y_L. The candidate-variable center is c=y_L-e_0=P^-1 T y_L. The gap is the squared P-distance of y_d from c minus rho squared. The ideal match sits on the ellipsoid boundary. Points inside beat it, points outside lose to it, and boundary points tie.

**Explanation / transition:** By P-inner-product Cauchy–Schwarz the gap is at least ||e||_P(||e||_P-2rho). Therefore separation alone suffices when every mismatch exceeds twice rho. S is SPD by the Woodbury identity; T is PSD, and is SPD for gamma>0 using the full row rank of A. The characterization requires the same positive regularizer and candidate-independent coupling.




**Slide 49. Purpose / pace:** Question: what about noisy imperfect matches?

**Talking points:** The expansion of a quadratic is exact: cost change is the gradient dotted with xi plus xi transpose P xi. Bound its magnitude using ||P|| <= 1/2. Apply the uniform eta to all candidates, including the formerly ideal one, and use the elementary margin theorem.

**Explanation / transition:** The matrix bound follows from T <= A K^-1 A^T/4 and S=(I-A K^-1 A^T)/2, hence P <= I/2. This is an observation-level result for a fixed unweighted model, not already a bound for adaptive weights, recomputed normalization, changing candidate windows, or the final aggregated volume. The general finite-list margin theorem applies once an appropriate score perturbation bound is independently established.

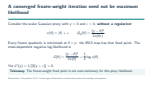


**Slide 50. Purpose / pace:** Question: where does the contraction constant come from?

**Talking points:** The difference identity isolates how changing weights changes the exact minimizer. For X=B_time^T W^1/2, M_W >= XX^T+ell I gives ||M_W^-1X|| <= ell^-1/2. The residual bound comes from comparing the frozen optimum with zero. The reciprocal identity uses ||s|-|s'|| <= |s-s'| and two bounded reciprocal weights.

**Explanation / transition:** This produces the epsilon^-3/2 factor after applying the weighted residual bound, then ||s-s'||_infty <= ||B_time|| ||g-g'|| and ||B_time||=||A||. If $q \geq 1$, the bound is inconclusive. At reported $n = 20$, $\varepsilon = 0.5$, and $\lambda \in [0.01, 0.1]$, the sufficient threshold $\|\mathbf{y}\| < \lambda\varepsilon^{3/2}/[2\cos(\pi/40)]$ ranges from about 0.001773 to 0.017732. Actual candidate norms are unavailable, so no experimental $q$ is certified. This concerns the unclipped map.

A converged frozen-weight iteration need not be maximum likelihood




**Slide 51. Purpose / pace:** Question: does IRLS recover a likelihood optimum?

**Talking points:** At fixed variance, the log-variance term is constant and the frozen quadratic derivative is just (theta-y)/v_k. If variance is state-dependent, differentiating the likelihood also differentiates the variance and its logarithm. At theta=y the residual terms vanish but the log-variance derivative remains positive.

**Explanation / transition:** This simplest counterexample establishes the logical gap without depending on the actual temporal difference matrix. The dissertation also supplies an example with A=[-1,1] and positive regularization where the fixed point is not stationary for the final-state-weighted score, even without the log term. Performance of the surrogate can still be evaluated empirically.




**Slide 52. Purpose / pace:** Question: what exactly do the experiments run?

**Talking points:** The implementation builds clipped solve weights, but scores at the new state with unclipped recomputed weights. A ratio cap or cost-change rule may stop the procedure before reaching a fixed point. Therefore the returned latent state is an iterate, not a certified minimizer of one state-dependent objective.

**Explanation / transition:** Candidate guidance and uniqueness rejection insert finite penalty values before spatial averaging and SGM. Averaging mixes them with genuine costs, so a soft guidance window is not a hard feasibility guarantee for the final winner. WTA is exact over its computed list, but semi-global path aggregation is not claimed to solve a global two-dimensional disparity objective exactly.

Depth evaluation distinguishes returned coverage and scored coverage


Depth evaluation distinguishes returned coverage and scored coverage

1. Issued both a false and a true reporting-depth band.
2. The returned estimate is false.
3. The estimate also lies in the reporting band.

| configuration | Returned coverage | Scored coverage | Scored points |
|---|---|---|---|
| Control | 96.97% | 70.36% | 1,447 M |
| WLS | 96.49% | 70.44% | 1,449 M |

This panel fixes $d_{\mathrm{rep}} = 0.5$ ft. True-shared-false depths are excluded by construction; they are not


**Slide 53. Purpose / pace:** Question: is WLS simply much denser?

**Talking points:** Its returned-estimate coverage is similar to control at the reported aggregate level, and the small difference changes sign across sequences. The main scored-point difference comes from much higher estimate-side in-band retention. The benchmark excludes out-of-band estimates, so an increase in scored points cannot be described as a general increase in every kind of coverage.

**Explanation / transition:** The two complete configurations differ in more than weighting, including final activity masking. The aggregate percentages use the dissertation's conventions. Raw outputs here exclude the separate posterior refinement operator but include the finest-level interpolation of the score minimum. Accuracy and retained coverage should be assessed jointly, ideally on a common population with out-of-band behavior also reported.

Pooled kurtosis does not establish heavy-tailed active errors




**Slide 54. Purpose / pace:** Question: why not infer a heavy-tailed likelihood from the residual plot?

**Talking points:** A large pooled kurtosis can be produced by many exact zeros even when the nonzero component is Gaussian. The moment calculation makes this clear. Fitted residuals also need not inherit the independence or equal variance of observation errors after regularized fitting.

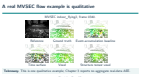
**Explanation / transition:** This motivates an event-active analysis reporting the zero fraction, fitted-residual variance, and sequence dependence, ideally compared with synthetic residuals generated under the proposed observation model through the same fitting procedure. Robust bounded-influence penalties remain possible modeling choices, but their quantitative benefit is not established by the pooled plot alone.




**Slide 55. Purpose / pace:** Question: what do the adaptive weights do?

**Talking points:** Chapter 2 lowers the magnitude penalty where event activity is high. Chapter 4's empirical rule increases its weight with pre-normalization energy. The rules answer different design choices; no shared optimality theorem explains their opposite directions.

**Explanation / transition:** For differentiable parts of the Chapter 2 rules, perturbing the normal equations gives $\delta\mathbf{f}^* = \mathbf{K}^{-1}(\mathbf{A}^\top \delta\mathbf{y} - \delta(\mathbf{\Lambda}^2)\mathbf{f}^*)$. Absolute values and clipping can have kinks, and no full pipeline Lipschitz result is established. The frame-assisted bilevel comparator also receives an image input, unlike event-only reconstruction.

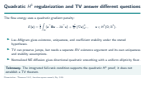



**Slide 56. Purpose / pace:** Question: what is the real-data AEE?

**Talking points:** The dissertation does not report an aggregate real-data AEE for this comparison. State that directly; no value is available in the reported results.

**Explanation / transition:** The event-accumulation baseline is a separate comparison method. The theorem's coefficient representations are the time-surface, voxel, and structure-tensor constructions. These panels support a qualitative comparison only and establish no numerical ranking.

## Quadratic $H^1$ regularization and TV answer different questions

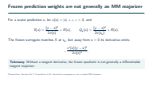



**Slide 57. Purpose / pace:** Question: why use $H^1$ instead of TV?

**Talking points:** The quadratic model yields a bilinear form to which Lax–Milgram applies, then an energy identity gives coefficient stability. TV may retain discontinuities, but requires a separate existence framework in $BV$.

**Explanation / transition:** A residual-square fidelity may be considered for a TV model, but no such theorem is established here. Replacing the gradient term with TV in an abstract signed-linear energy can make it unbounded below. Full-rank pointwise fidelity can retain strict convexity, so TV does not always lose uniqueness; the integrated condition alone no longer supplies $H^1$ coercivity.

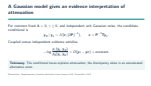
Frozen prediction weights are not generally an MM majorizer




**Slide 58. Purpose / pace:** Question: is this IRLS a majorization-minimization method?

**Talking points:** In general, no. The state-dependent variance contributes the displayed derivative term, which the frozen-weight quadratic omits. Thus matching the objective value at the current iterate does not give a differentiable tangent majorizer.

**Explanation / transition:** No generic descent guarantee follows. The proved existence and conditional contraction results concern a fixed map. The implementation's cost-rise acceptance rule is separate and can allow small rises, so do not claim monotone cost decrease.




**Slide 59. Purpose / pace:** Question: can attenuation be read probabilistically?

**Talking points:** For fixed common parameters, interpret the magnitude and coupling penalties as a Gaussian prior and observations as independent unit-noise Gaussian measurements. The conditional distribution makes the profile prediction and its uncertainty explicit.

**Explanation / transition:** The prior precision is $\mathbf{M}_0 = \begin{pmatrix} \mathbf{\Lambda}^2+\gamma I & -\gamma I \\ -\gamma I & \mathbf{\Lambda}^2+\gamma I \end{pmatrix}$, and $\mathbf{M} = \mathbf{M}_0 + \operatorname{diag}(\mathbf{A}, \mathbf{A})^\top \operatorname{diag}(\mathbf{A}, \mathbf{A})$. Integrating out the profiles gives $-\log p_\gamma(\mathbf{y}) = C_\gamma(\mathbf{y}) + \frac{1}{2}\log\det\mathbf{M} - \frac{1}{2}\log\det\mathbf{M}_0 + \text{constant}$; these determinant terms are fixed across candidates. The operator $\mathbf{P}^{-1}\mathbf{T}$ has eigenvalues in $[0, 1)$ and is self-adjoint and contractive in the $\mathbf{P}$ metric, but need not contract in Euclidean norm. For scalar $\mathbf{\Lambda}$, attenuation is diagonal in the temporal observation eigenmodes. The criterion from $D$ alone selects an ideal match for $\gamma > 0$, full-row-rank $\mathbf{A}$, and distinct candidate profiles; it is an unevaluated score alternative, not a disparity MAP result or evaluated method.

The implemented estimator has several distinct layers




**Slide 60. Purpose / pace:** Implementation scope; 60 seconds.

**Talking points:** Expand IRLS as iteratively reweighted least squares and SGM as semi-global matching: SGM penalizes disparity changes along multiple paths to encourage spatial coherence. A finite search exactly returns a minimum of the final score list, subject to tie breaking. That narrow exactness should not be confused with correctness of the physical correspondence or exact optimization of a global two-dimensional labeling energy.

**Explanation / transition:** Clipping affects solve weights; final scores recompute unclipped weights. Finite penalty sentinels are averaged before selection, so the guidance window is soft. The benchmark includes the finest-level interpolation of the aggregated score but excludes the separate hole-filling/refinement operator. The further analysis distinguishes the optimization and scoring steps explicitly.




**Slide 61. Purpose / pace:** Question: what is new if the proof tools are classical?

**Talking points:** Do not claim to invent Tikhonov regularization, Horn–Schunck, Lax–Milgram, or Brouwer. Explain the work as constructing and comparing precise event-based inverse models, identifying what assumptions make them well posed, and deriving consequences that are otherwise easy to overclaim. The common flow theorem gives a reusable treatment instead of separate incomplete proofs.

**Explanation / transition:** The stereo cost decomposition and attenuation example explain why inner solvability does not guarantee correct correspondence. The separation and margin conditions supply a conditional answer. The fixed/adaptive split prevents a quadratic guarantee from being incorrectly transferred to a changing-weight procedure. These are concrete mathematical insights that inform model design and evaluation.




**Slide 62. Purpose / pace:** Attribution.

**Talking points:** The reconstruction references distinguish learned event-only video reconstruction, flow-based image inversion, and frame-assisted bilevel threshold estimation. The flow references identify the classical variational structure, timestamp geometry, existing well-posedness theory, and nonconvex contrast maximization.

**Explanation / transition:** The dissertation's contribution is the stated event-derived models, shared coefficient analysis, and derived conditioning, bias, and selection consequences. The presentation's empirical values and theorem statements come from the dissertation. The source map ties each slide to its relevant labels, figures, and tables.

Stable estimates can still have regularization bias




**Slide 63. Purpose / pace:** Question: does stability imply truth?

**Talking points:** No. The estimator's error splits into a shrinkage bias and propagated noise. The stability bound controls sensitivity to the observations, while the bias depends on the signal and the penalty.

**Explanation / transition:** A zero-mean reference fixes the unobservable constant ambiguity. On a fixed grid, noiseless data from a mean-zero reference signal are recovered as $\lambda \to 0$. Persistent noise can prevent recovery as $\lambda \to 0$; uniform refinement requires controlling noise, regularization, and grid scaling. Model error may enter the residual separately from sensor noise.

The physical event stream loses more than one baseline




**Slide 64. Purpose / pace:** Question: is the constant nullspace the whole sensor ambiguity?

**Talking points:** No. The chain matrix has exactly the constant nullspace, but the sensor only fires after a contrast threshold crossing. The two displayed positive intensity signals share the same initial reference and remain inside that threshold, so both produce the same empty event stream.

**Explanation / transition:** This example shows that a physical stream can omit changes that never cross threshold. It does not say arbitrary perturbations preserve a nonempty event schedule. A small perturbation supported strictly away from existing crossings can preserve that schedule when there is a positive threshold margin.


Integrated directional information is necessary for uniqueness

Let $\Omega$ be connected, bounded, and Lipschitz; $n \in \mathbb{N}$, $\mathbf{B} \in L^\infty$ symmetric PSD, and $\kappa \in \mathbb{R}$. Minimize over symmetrical $H^1(\Omega, \mathbb{R}^n)$ fields: $\text{Tot. Eq.} \int_\Omega \|\mathbf{B} \cdot \nabla \mathbf{A}\|_F$ is singular, choose $\alpha \neq 0$ with $\mathbf{B}\alpha = 0$. Then $\mathbf{B}\alpha = 0$ a.e. and

$$E[\mathbf{a} + \kappa \alpha] < E[\mathbf{a}] - \kappa c^2 \int_\Omega \mathbf{a}$$

If the coefficient is nonzero, $E$ is unbounded below; if zero, it is flat in direction $\alpha$.

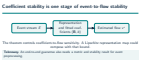
**Takeaway.** The integrated matrix condition is necessary for a unique minimizer in the unconstrained quadratic model.

Connections: Theorem 3.1, which extends this to the nonlinear setting.

**Slide 65. Purpose / pace:** Question: why require $\int \mathbf{B} \succ 0$?

**Talking points:** A singular integrated matrix leaves a nonzero constant flow direction unseen by both the data quadratic and the gradient penalty. Along that direction, the remaining linear term either drives the energy without a lower bound or vanishes.

**Explanation / transition:** Because $\mathbf{B}$ is PSD, $c^\top \mathbf{B}(x)c \geq 0$ a.e.; its integral is $c^\top M_B c = 0$, so $\mathbf{B}^{1/2}c = 0$ and hence $\mathbf{B}c = 0$ a.e. In the zero-slope case, any existing minimizer would have a flat family, so uniqueness fails; this alone does not prove a minimizer exists. Residual-square data can make the linear coefficient compatible with the null direction. Boundary constraints or a ridge term change the argument.




**Slide 66. Purpose / pace:** Question: how stable is the full event-to-flow pipeline?

**Talking points:** The variational theorem begins after events have become coefficient fields. If preprocessing were Lipschitz in a chosen event metric, its bound could compose with the coefficient-to-flow estimate.

**Explanation / transition:** The metric matters: moving one positive unit atom by nonzero $\delta t$ gives the total variation norm $\|\delta_{t+\delta t} - \delta_t\|_{\mathrm{TV}} = 2$, while the single-atom mass-one Wasserstein distance is $|\delta t|$. Hard bins or changing candidate neighborhoods may jump; fixed-bandwidth smoothing can behave better under a suitable metric. No global pipeline theorem is established here.



**Slide 67. Purpose / pace:** Question: should $\lambda$ change as the grid is refined?

**Talking points:** There is no answer without naming a target continuum functional. In this smooth uniform-time illustration, difference residuals scale like $h$, while the Riemann sum contributes $1/h$. Matching the magnitude penalty yields $\lambda_h = \beta_{\mathrm{time}} h$.

**Explanation / transition:** Finite-endpoint quadrature may need endpoint weights, and this calculation is not a convergence theorem. A physical event model would also require a scaling choice for noise and kernel bandwidth. It does not assert that the dissertation's finite-grid experiments must use this law.

**Slide 68. Purpose / pace:** Question: what happens as $\alpha$ grows?

**Talking points:** The gradient penalty forces the minimizer toward a constant field. The data term then selects the constant that balances the integrated directional information against the integrated forcing.

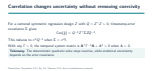
**Explanation / transition:** For a proof, write $w_\alpha = v_\alpha - c$ and test the weak equation with it. With $h_b = b - \mathbf{B}c$ and $\int h_b = 0$, the identity is $\int w^\top \mathbf{B} w + \alpha^2 \|\nabla w\|_2^2 = \int h_b \cdot (w - \bar{w})$. Poincare gives $\|\nabla w\| = O(\alpha^{-2})$. Testing constants yields $\int \mathbf{B} w = 0$; positive definiteness of $\int \mathbf{B}$ and Poincare then give $\bar{w} \to 0$. This is a direct corollary, not a separately stated theorem. No blanket $\alpha \to 0$ claim follows.




**Slide 69. Purpose / pace:** Question: should events be modeled as a conserved fluid?

**Talking points:** That is an additional physical assumption. Scalar transport follows a pattern value along a trajectory; density conservation also accounts for compression or expansion through the divergence of flow.

**Explanation / transition:** For a simple illustration, in one dimension let $v(x) = x$ and $V(x,t) = 1$. It solves scalar transport, but source-free conservation has residual $\partial_x(Vv) = 1$. A source term $V_t + \nabla \cdot (Vv) = s$ may be appropriate if events are created or lost, but this dissertation establishes no physical event conservation law.




**Slide 70. Purpose / pace:** Question: what if timestamp errors are correlated?

**Talking points:** Independence is not needed for the covariance formula of a linear OLS estimator; the full covariance matrix replaces the white-noise simplification. If each error has the same mean, the centered symmetric design cancels that mean in the slope.

**Explanation / transition:** Smoothing errors with $H_\epsilon$ produces $H_\epsilon \Sigma H_\epsilon^\top$, which may be singular, so do not invert it without checking. A positive definite fidelity precision $\Gamma^{-1}$ keeps the regularized normal matrix positive definite, but changes coefficients and constants. A Gaussian likelihood interpretation needs a separate distributional assumption; the deterministic theorem does not validate a noise model.




**Slide 71. Purpose / pace:** Divider for questions.

**Talking points:** Use when asked how time-surface gradients are estimated, or about contrast maximization, the most widely used event-based motion objective.

**Explanation / transition:** Contrast maximization answer in one sentence: a global minimizer exists by finite-dimensional coercivity with the coefficient ridge; the local solver carries no global guarantee.



**Slide 72. Purpose / pace:** Gradient-estimation result; 75 seconds; shorten to the takeaway in a compressed talk.

**Talking points:** The design matrix Q depends on sample positions, not the observed slope. Central symmetry decouples the intercept and cancels the quadratic Taylor term in the slope. A Lipschitz Hessian leaves a cubic remainder bounded by L_H |z|^3/6. Multiplication by z and Q^-1 yields the bias bound. Independent equal-variance errors give the covariance, and the MSE decomposition combines both contributions for the same discrete estimator.

**Explanation / transition:** A full-rank disk of timestamps around a moving straight edge can estimate its gradient well while the tangential velocity remains unidentified. The assumed timestamp-noise model is not derived from event-count statistics. The asymptotic radius heuristic is in the backup, with its finite-design qualifications.

Contrast maximization finds the motion that sharpens the events




**Slide 73. Purpose / pace:** What CMAX models and why it is promising; 60 seconds.

**Talking points:** Describe the warp in scene terms: the right motion turns a smeared trail into a sharp edge. The quadratic models first compress events into a time surface or bins and fit coefficients; CMAX keeps each event's own time. In all three known-motion scenes its AEE is the lowest, by 0.66 to 1.66 px over the structure tensor. On the stripes it recovers the diagonal, where interior pixels alone determine only the across-stripe component (normal-flow reference error 2.12 px).

**Explanation / transition:** Scope: one representative window (chosen by the median-deviation rule) on simulated scenes with known motion. On the idealized stripes CMAX also assigns flow to the surrounding background. The dissertation reports no aggregate real-data AEE for any flow method. Its guarantee is existence of a global minimizer; the quasi-Newton solver is local. The next two backups give the objective and the existence proof.




**Slide 74. Purpose / pace:** Nonconvex extension; 60 seconds.

**Talking points:** In this model the unknown changes the represented data during optimization. q is displacement in pixels per normalized window, avoiding a confusion between velocity and normalized time. A continuous spline basis supplies well-defined event-location evaluations. General H1 equivalence classes in two dimensions cannot be evaluated at prescribed points without more regularity.

**Explanation / transition:** The regularizer has a diffusion floor and a coefficient ridge with different roles. A positive diffusion floor penalizes all nonconstant spline variations but still leaves global translations free. The coefficient ridge controls those constant-flow directions and supplies coercivity of the full finite-dimensional objective.

## A global minimizer exists, even though the solver is local



**Slide 75. Purpose / pace:** Existence proof; 75 seconds.

**Talking points:** A bilinear tent contribution is a continuous function of the warped event location, including when a location crosses a pixel boundary. Each event distributes at most its absolute weight onto the canvas, so the image L1 norm and maximum absolute entry are at most W_tot. Bound variance by the second moment, then by max times L1 divided by the number of pixels.

**Explanation / transition:** The negative focus term is bounded below. The positive coefficient ridge drives the energy to infinity as the coefficient norm grows, so a global minimizer exists by Weierstrass. This is a finite-dimensional existence result; uniqueness and convergence of the local solver remain open. The evaluated tent objective is only piecewise differentiable, and the returned solution is not certified global. The ridge bound $2\varepsilon/M$ is not uniform under refinement.

Stereo depth: full Chapter 4 arc

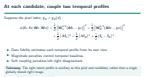



**Slide 76. Purpose / pace:** Divider for questions; the full stereo arc runs about 17 minutes if presented in sequence.

**Talking points:** Choose the page that matches the question. For “does the correct disparity win?”, go to the counterexample and then the selection theorem. For “does IRLS converge?”, go to the fixed-point slide and its backup proof.

**Explanation / transition:** The summary slide in the core states the four results these pages prove.

At each candidate, couple two temporal profiles




**Slide 77. Purpose / pace:** Stereo model; 75 seconds.

**Talking points:** The temporal operator is exactly the first-difference matrix used earlier. Lambda is a fixed positive diagonal regularizer during the candidate's iteration, and gamma is nonnegative. The right observation varies with disparity. The right latent unknown is solved anew for every left pixel and candidate. Spatial coupling enters later through the candidate-cost aggregation.

**Explanation / transition:** Keep the semantic distinction clear: these are uncalibrated event-feature profiles because warping, downsampling, and affine normalization transform the observations. Calling them absolute log-intensities would require a separate calibrated observation model. Unweighted theory is the case W_L=W_R=I.



**Slide 78. Purpose / pace:** Stereo SPD proof; 90 seconds.

**Talking points:** The proof reuses the temporal argument, now with an additional nonnegative coupling term. The Hessian is M_W. Every direction receives the magnitude-penalty lower bound, regardless of whether the data provide a strong temporal signal. This makes each frozen system invertible.

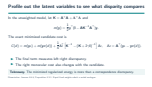
**Explanation / transition:** With fixed weights, subtracting normal equations gives stable dependence on the data, governed quantitatively by the inverse matrix norm and conditioning. This says nothing yet about identifying a unique disparity. Nor does it mean that updating weights solves the same strictly convex function.




**Slide 79. Purpose / pace:** Spectral structure; 90 seconds.

**Talking points:** Adding the two normal equations eliminates coupling; subtracting them adds 2 gamma to the temporal precision. The orthonormal transform makes this exact and preserves Euclidean condition numbers. For scalar regularization, K has smallest eigenvalue lambda squared and largest g_max+lambda squared, so the displayed formula is exact.

**Explanation / transition:** Coupling increases the largest curvature but does not repair the common constant baseline mode, so the full-space condition number rises with gamma at fixed lambda. Unequal weights do not admit this simple block diagonalization. The mean-zero restriction avoids the baseline mode for structured perturbations, but full-space conditioning remains relevant to unstructured numerical errors; see backup.

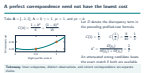


**Slide 80. Purpose / pace:** Profiled cost; 90 seconds.

**Talking points:** Complete the square in the stacked latent variables: the minimum equals ||y|| squared over two minus h^T M^-1 h over two. Use the sum–difference transform and the parallelogram identity to separate the independent monocular costs from the coupling discrepancy. The discrepancy matrix is PSD because the inverse difference is 2 gamma K^-1(K+2 gamma I)^-1.

**Explanation / transition:** An exact winner of these unweighted, unaggregated profiled costs also minimizes the corresponding joint latent/disparity objective; it has no physical-correctness guarantee. For fixed common $\mathbf{\Lambda}, \gamma$ and unit observation noise, an ideal Gaussian model with latent prior precision $\mathbf{M}_0 = \begin{pmatrix} \mathbf{\Lambda}^2+\gamma I & -\gamma I \\ -\gamma I & \mathbf{\Lambda}^2+\gamma I \end{pmatrix}$ gives profiled $C_\gamma(y)$ as $-\log p_\gamma(y)$ up to a candidate-independent constant. Its linear conditional prediction is $\mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L$; scalar-$\mathbf{\Lambda}$ modes are attenuated. This is not disparity MAP or a consistency guarantee.

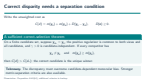
A perfect correspondence need not have the lowest cost




**Slide 81. Purpose / pace:** Counterexample; 90 seconds.

**Talking points:** The example has just two temporal latent samples, so one signed difference observation per view. For each b the objective is a strictly convex quadratic and is minimized exactly. Nevertheless the perfect right observation b=1 has greater cost than b=1/2 because attenuation reduces the right monocular penalty enough to offset the discrepancy.

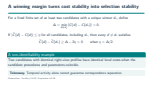
**Explanation / transition:** The result does not say the evaluated weighted algorithm always picks an attenuated match. It is a rigorous counterexample to deducing correct correspondence from the unweighted inner convexity theorem. In general C(b)=(1+b^2)m(y_L)+(1-b)^2 D(y_L), so C'(1)=2m(y_L)>0 for a nonzero observation and positive regularization.




**Slide 82. Purpose / pace:** Selection theorem; 75 seconds.

**Talking points:** Under gamma>0 and full row rank of the temporal difference matrix, the discrepancy quadratic D is positive definite on the observation space. Subtract C(d*)=2m(y_L) from C(d). The monocular difference is nonnegative by assumption, and D is strictly positive for a distinct observation. Every competitor therefore loses.

**Explanation / transition:** This is sufficient, not necessary. The theorem's second separation criterion handles competitors with lower monocular costs if they are far enough away in the induced metric. Both statements assume ideal equal profiles at the physical match and concern unweighted candidate-local costs before masking or aggregation. They do not assert these conditions hold on the experimental sequences.




**Slide 83. Purpose / pace:** Stability and ambiguity; 75 seconds.

**Talking points:** A competitor's score may move down by eta while the winner's moves up by eta, so half the original margin is the sharp elementary threshold for this bound. The observation-based specialization is in the backup. This bridges continuous coefficient/data stability and the discrete argmin decision.

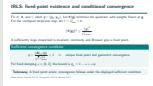
**Explanation / transition:** For fixed parameters, the negative log coupled-versus-independent Gaussian likelihood ratio equals the discrepancy $D = C_\gamma - C_0$ up to candidate-independent determinant terms. The discrepancy $D$ alone favors an ideal identical profile over distinct competitors when $\gamma > 0$ and $A$ has full row rank; this is an unevaluated alternative comparison, not a new result here. Coincident right profiles still tie under identical local procedures, though spatial aggregation can create contextual differences; keep this local observation scoped to local costs.



**Slide 84. Purpose / pace:** Weight motivation; 75 seconds.

**Talking points:** The difference of independent Poisson counts is Skellam. In a high-count regime, a Gaussian approximation motivates weighted least squares with an inverse-variance weight. But the raw variance is the sum of polarity rates while the mean is their difference. A net-rate-dependent variance is an added approximation, not a consequence of Skellam moments.

**Explanation / transition:** Furthermore the actual observations are real signed masses after warp, downsampling, and normalization, not untouched integer Poisson counts. The weighting proxy is evaluated at the preceding state and then frozen. Its reciprocal form downweights high predicted activity relative to low activity. The reported comparison supports this rule on method-specific scored populations.



**Slide 85. Purpose / pace:** IRLS analysis; 90 seconds.

**Talking points:** Compare the frozen minimizer's objective value with the value at zero. The regularizer bounds the squared norm from below, while reciprocal weights are uniformly bounded above by 1/epsilon. Thus every closed ball with radius $R \geq \|\mathbf{y}\| / \sqrt{\ell \varepsilon_{\text{noise}}}$ is invariant. The weights are continuous and frozen matrices remain SPD, so $\Phi$ is continuous. Brouwer proves a fixed point.

**Explanation / transition:** A sharper difference-of-normal-equations argument gives the Lipschitz constant q. If q<1, Banach supplies uniqueness and convergence. The convergence conclusion applies under this sufficient condition; the evaluated stopping and acceptance rules are additional steps. Damping preserves fixed points, but the stated upper bound does not prove damping helps when q>=1. A fixed point need not minimize the state-dependent score or a Gaussian likelihood.

## Stereo benchmark: read accuracy alongside scored coverage


Stereo benchmark: read accuracy alongside scored coverage
MAVEC indoor flying, raw outputs, including Scene-level cases interpretable
Mean depth error (cm) log RMSE Scored
Method Fly Fly Fly Fly ×100 ↓ points (M)
Hirschmüller (ref.) 549.55 615.46 542.35 613.64 56.16 3.42
Piatkowska (local) 54.03 54.03 54.03 54.03 54.03 3.42
Ours (coupled WLS) 73.63 58.55 70.15 54.25 49.50 3.66
Piatkowska (dissertation) 67.92 55.75 60.46 54.63 49.72 3.66
Each method is scored on its own finite estimates and ground truth in [0.5, 6] m. Point counts are
Takeaway: WLS is competitive with a major overall population. The table does not compare
accuracy as a common yard set.
Source/note: Table 5.6. Data for overview and not in dissertation.

**Slide 86. Purpose / pace:** Primary stereo results; 90 seconds.

**Talking points:** Introduce only the columns shown. Mean absolute depth error is in centimeters separately for each sequence. The displayed log RMSE is one hundred times the base metric, then averaged equally over sequence-level values. Point counts sum scored pixels across sequences. Piatkowska's local adaptation leads on most error metrics in the full dissertation table; WLS leads on Flying 2 and narrowly on log RMSE.

**Explanation / transition:** The valid set differs by method. Both estimates and ground truth must be in the reporting depth band. A method can return many finite depths but have fewer in-band scored points. Complete configurations also differ in normalization, regularization, guidance, and final activity masking. Interpret the reported accuracy together with scored coverage and the complete method configuration.

## A weighting comparison supports the selected rule




**Slide 87. Purpose / pace:** Stronger empirical evidence; 60 seconds.

**Talking points:** The stated configurations vary the weighting rule while keeping the remaining settings fixed. Each method is scored on its own valid population. The displayed differences subtract the WLS mean errors from the WLS-Prop mean errors in the benchmark table.

**Explanation / transition:** The reciprocal rule was chosen using these sequences, so it is not a held-out validation result. No claim of maximum likelihood follows from empirical performance. A later controlled experiment could compare against truly homoscedastic weights under the same preprocessing, regularizer, and validity rules.

Hard fusion changes both the constraint and the regularization




**Slide 88. Purpose / pace:** Fusion result; 75 seconds.

**Talking points:** Set the two profiles equal in the soft energy: its two magnitude penalties add to ||Lambda v|| squared. The implemented JIF uses only half that penalty. Hence its reduced system is not simply the infinite-gamma limit with unchanged regularization. Divide its system by two to expose the effective half squared regularizer.

**Explanation / transition:** The mean of the soft model is already a fused profile, as the sum equation shows. Therefore saying that averaging destroys contrast is not an established explanation for the JIF performance gap. The experiments compare complete configurations with different shared weights, maximum iteration counts, and scored populations. The counterfactual effect of hard equality alone remains to be isolated.




**Slide 89. Purpose / pace:** Formal aperture model; about two minutes if asked.

**Talking points:** Here L is log-intensity. The profile ell is translated across a straight edge, with a fixed unit normal and steady illumination. Substitution into brightness constancy gives ell prime times the quantity n dot v minus s. Wherever the profile changes spatially, this determines only the normal component. Adding any constant tangential velocity produces exactly the same log-intensity history at each pixel, and therefore the same ideal threshold events from identical reset references.

**Explanation / transition:** For zero normal speed, translation is entirely along the ideal edge and the pixel values stay constant. This local model assumes endpoints and other distinguishing features lie outside the observation window. For nonzero normal speed, the crossing-time map has gradient n divided by s, which gives the time-surface constraint. Illumination changes are a separate source of temporal change covered by the original threshold rule.